

HW: Introduction to Machine Learning

In below, I will discuss the step-by-step approach to make Data Analytic flow to create a Machine Learning predictive model. You can use this flow as a template to solve any supervised ML and apply it to your project with the data you have been assigned.

The flow (Key steps) of the case study is as below:

1. Defining the problem statement

- Data description
- Defining the problem (in term of general problem and ML problem)
- What is the **Target Variable** ?
- What is the **Predictors (features)** ?
- What is the definition of your **Samples**?
- **Note:** If you have assumptions for designing Target Variables and Predictors, you can be further explained in this section.

2. Looking at the distribution of Target variable and describe it.

- Balance or not ?
- Skewness ?
- Outlier treatment should be done ?

3. Basic data exploration for rejecting useless columns.

- This step is performed to gauge the overall data. The volume of data, the types of columns present in the data. Initial assessment of the data should be done to identify which columns are **Quantitative, Categorical or Qualitative**.
- This step helps to start the column rejection process. You must look at each column carefully and ask, does this column affect the values of the Target variable? If the answer is a clear "No", then remove the column immediately from the data, otherwise keep the column for further analysis.

4. Visual Exploratory Data Analysis for data distribution and feature selection based on data distribution.

- Categorical variables
 - **Guidelines:** We can spot a categorical variable in the data by looking at the unique values in them. Typically a categorical variable contains less than 20 Unique values AND there is repetition of values, which means the data can be grouped by those unique values.

- Balance or not ?
- Skewness ?
- Outlier treatment should be done ?
- Etc.
- Continuous variables
 - **Guidelines:** Histograms shows us the data distribution for a single continuous variable. If there is too much skewness, then outlier treatment should be done and the column should be re-examined, if that also does not solve the problem then only reject the column.
 - Balance or not ?
 - Skewness ?
 - Outlier treatment should be done ?
 - Etc.

5. Outlier treatment

- Summary feature columns that you done with outlier treatment and visualize all features again.

6. Missing Values treatment

- Missing values are treated for each column separately.
- If a column has more than 30% data missing, then missing value treatment cannot be done. That column must be rejected because too much information is missing.
- Delete the missing value rows if there are only few records.
- Impute the missing values with MEAN, MEDIAN, MODE value or others. Why ?
- Interpolate the values based on business logic (You can use the resources available on the internet to manage your missing values. But please cite the source.)

7. Feature selection by Visual correlation analysis and Statistical measurement

- Now its time to finally choose the best columns (Features) which are correlated to the Target variable. This can be done directly by measuring the correlation values, ANOVA/Chi-Square tests or others. However, it is always helpful to visualize the relation between the Target variable and each of the predictors to get a better sense of data.

- Visual exploration of relationship between variables
- Statistical measurement of relationship strength between variables

8. Model Construction

- Data Pre-processing for Machine Learning (Converting data, Splitting data, Normalization, etc.)
- Model Building (HW1-linear regression, HW2-logistic regression, HW3 others)
- Results analysis

9. Conclusion (Examples)

- How accurate is each model? and which model gives the most accuracy?
- Overfitting or not? Why?
- Based on the accuracy you get from the models, can your model actually be used in real situation?
- What features are important and sufficient to create a ML model